Financial Data Analysis Using Singular Value Decomposition (SVD)

CS685: Foundations of Data Science

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1. Overview:

- **Problem statement:**

  The historical stock data is very difficult and hard to understand due to its random changes in the prices of increasing or decreasing and also it is very complicated to predict their future trends. There are various factors included in the stock data like open, close, low and high which are recorded over long periods, for the traditional methods it is very complex to find out the clear and also patterns which are useful. Some of the factors make it more important like Data complexity, Noise reduction and Prediction.

  1. Data complexity: As the stock data is very huge it has multiple features, due to those features it is difficult to analyze the data.
  2. Noise reduction: Important trends can be hidden due to the random changes in the stock data.
  3. Prediction: It is very important to concentrate on useful patterns and ignoring the noise to make the correct and accurate forecasts results.

  The main goal of understanding basic trends in the stock market data are:

  1. Finding the behavior of the market.
  2. Improving the trading strategies.
  3. Using better and accurate models for predicting the random fluctuations of prices.

- **How SVD Solves It:**

  SVD: Singular Value Decomposition is one of the mathematical techniques which is used to split the complex matrix into the simpler ones. This decomposition of data is a good idea as it helps us to understand each and every part of data very clearly and effectively and also this technique is widely used in booming fields in today's era like machine learning and financial data analysis.

  There are few reasons why SVD is essential in solving the financial data analysis:

  1. Simplification of data:

There will be a lot of unwanted details in the financial data which are less likely used, for this SVD is responsible for filtering the data and seems to focus on the important trends and in parallel ignoring the noise.

2. Identifying the key patterns:
   SVD filters individual pieces of data, which is very useful for finding the main patterns which are responsible for the change in the stock prices. These founded patterns help to understand the data and use it for data forecasting.
3. Deleting Noise:
   SVD will remove the unwanted details from the stock and filters the majorly used important patterns of data.
4. Trend visualization:
   Visibility is the important factor to analyze the trends, here SVD will makes it very easy to see and understand the trend fluctuations of data, based on that we can take wise decisions to improve the stock market prices.
5. Future prediction:
   For forecasting in short-terms SVD will recognize the trends into the future as a building tool.
   Reasons for saying SVD serves as a great tool for financial data:
   As we know that SVD will remove the unwanted details and only shows the majorly required data which works as a compressing a very large image.
   When it comes to financial data it does:
   - Simplification in the complex nature of data: it just filters and concentrates on useful and wanted data.
   - Accuracy: reconstructing the data will few errors.
   - Improvement in visualization: it makes trends to see and understand easily.

By focusing on the major important parts of data, SVD will makes it simpler, clear and easier to understand and proceed to financial analysis and decision-making.

2. Methodology:

A. Proposed Methodology

1. **Data Collection:** We downloaded stock market information using the Yahoo Finance library (yfinance). In particular, the values of Goldman Sachs (GS) stocks were obtained between January 1, 2020, and November 1, 2024.
   The Open, High, Low, and Close prices were among the columns in the raw data. In order to comprehend the behavior of the stock, we concentrated our investigation on these characteristics.

2. **Data Normalization:** Since there can be significant fluctuations in stock prices, we normalized the data to facilitate pattern analysis. All features have the same scale because of normalization, which is computed as follows:

$$Normalized\ Value = \frac{Value - Mean}{Standard\ Deviation}$$

   Comparing trends and patterns across all stock price features—Open, High, Low, and Close—is made easier by this. The patterns were plotted over time using a normalized plot.

3. **Target Variable Creation:** For classification purposes, we defined the target variable as follows:
   1: The stock is in an upward trend if its close price is greater than its open price.
   0: If not, a decreasing trend is indicated.
   For classification tasks, this binary target variable is essential.

4. **Logistic Regression on original data:**
   Prior to using SVD, we assessed the classification performance using the original normalized data.

   train_test_split was used to divide the data into training (80%) and test (20%) groups.

To forecast the target variable, a Logistic Regression model was trained.

Metrics calculated:

Accuracy Score: The percentage of accurate forecasts.

F1 Score: Assesses categorization performance by combining recall and precision.

Initial performance:

Score for Accuracy: 95.08%

F1 Score: 95.59%

5. **Applying Singular Value Decomposition (SVD):**

SVD is a dimensionality reduction method that breaks down the matrix into three parts in order to find important trends in the data:

$$X = U \sum V^T$$

Where:

U: An orthogonal matrix that symbolizes "feature trends."

Σ: The significance of each trend is represented by a diagonal matrix of singular values.

V^T: The orthogonal matrix V^T is used to represent the "time-series contributions."

In order to capture the most important patterns, we chose three components for this analysis.

To minimize the dimensionality of the normalized data while keeping the most important information, the SVD technique was employed.

6. **Visualization of Trends:**

Using SVD, we generated the top three trends that were taken from the stock data. The most noteworthy patterns in stock prices over time are represented by these trends.

- Days are shown on the x-axis, and trend values are shown on the y-axis.
- One significant trend found by SVD is represented by each line.

7. **Matrix Reconstruction:**

We used only the top three components to recreate the original matrix in order to assess the quality of SVD. This technique entailed recombining the decreased trends into the original space.
By using a plot, we were able to compare the reconstructed and original data:

- Initial Close Prices
- Rebuilt Close Prices

These plots resemblance suggests that the essential information is adequately captured by the reduced components.

8. **Accuracy Metrics for Reconstruction:**

The following metrics were computed to assess the reconstruction's accuracy:

- Mean Squared Error: The average squared difference between the original and reconstructed data is measured by the Mean Squared Error, or MSE.

  MSE = $\frac{1}{n} \sum_{i=1}^{n} (x_i - \hat{x}_i)^2$

- Mean Absolute Error: The average absolute difference is measured by the mean absolute error, or MAE.

  MAE = $\frac{1}{n} \sum_{i=1}^{n} |x_i - \hat{x}_i|$

- R-squared (R^2): Indicates the percentage of variance that the reconstruction was able to capture.

$$R^2 = 1 - \frac{SS_{residual}}{SS_{total}}$$

Results:

MSE: 0.0001

MAE: 0.0054

R-squared: 0.9999

The remarkable accuracy of the reconstruction is confirmed by these results.

9. Logistic Regression on Reduced Data:

For categorization, the SVD-transformed reduced data was utilized. The same procedures were used as previously:

- Create training and test sets from the reduced data.
- Develop a model for logistic regression.
- Assess performance.

Post-SVD performance:

- **Accuracy Score:** 95.49%
- **F1 Score:** 95.94%

The little improvement shows that SVD reduces dimensionality while maintaining important information.

10. Forecasting Future Trends:

With the decreased data, we forecasted the future trends:

- To simulate future trends, the previous trend values were extended for 30 days.
- To illustrate possible future behavior, the predicted patterns were visualized alongside previous patterns.

11. Observations:

Trend Extraction: By eliminating noise and concentrating on significant trends, SVD efficiently finds important patterns in stock data.

Reconstruction Accuracy: The validity of the reduced components was confirmed by the minimal error in the reconstruction of the original data.

Classification Performance: After SVD, there was a small improvement in the performance of logistic regression on both original and reduced data.

Forecasting: The research offered a framework for extending current patterns into the future, which may provide information about how stocks may move in the future.

This methodology offers useful tools for financial data analysis and demonstrates the effectiveness of SVD in evaluating, recreating, and forecasting stock market movements.

3. Experiments

3.1 Setup:

- **Datasets:** The Datasets being utilized here are obtained from Yahoo Finance. All the datasets are fetched by **'yfinance'** python library and using the function '**yfinance.download()'.** The dataset consisted of approximately 1200 days records of the required stock prices. The dataset consists of various fields like date, open, close, low, high, and adj close. For this project, we have only considered one company stocks data i.e. **"Goldman Sachs Group Inc."** for data analysis and stimulation. We have only selected the features like open, close, high and volume for further analysis. These features have a strong correlation between them, as these capture the progression of movement in prices within a day.
- **Tools/Libraries Used:** We have used '**Python'** programming language for this project because of its libraries for analysis of data, visualization and machine learning.
  We have used various libraries for this project like:
  - numpy and pandas for handling data.
  - yfinance to download the data.
  - Matplotlib.pyplot for visualization of data.

- sklearn.decomposition.TruncatedSVD for applying SVD to reduce dimensionality.
- sklearn.metrics for error metrics.
- sklearn.model_selection.train_test_split for evaluating the performance.
- sklearn.linear_model.LogisticRegression for building simple classification model.

We have used Google Colab for the execution of this project.

➢ **Metrics for Evaluation**

1. Accuracy Score

Formula:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:

• TP: True Positive (predicted 1 and actual 1)

• TN: True Negative (predicted 0 and actual 0)

• FP: False Positive (predicted 1 but actual 0)

• FN: False Negative (predicted 0 but actual 1)

Explanation:

Accuracy represents the proportion of correct predictions (including both true positives and true negatives) out of the total predictions.

It is chosen for its simplicity and widespread use in classification tasks, providing an overall measure of how effectively the model predicts whether the closing price exceeds the opening price.

2. F1 Score:

Formula**:**

$$\text{F1 } Score = 2\frac{Precision * Recall}{Precision + Recall}$$

Where:

$$Precision = \frac{TP}{TP + FP} \qquad \text{Recall} = \frac{TP}{TP + FN}$$

Explanation:

Combines Precision (true positive rate) and Recall (sensitivity) into a single metric, making it particularly useful for imbalanced datasets.

Selected because stock trend classification often involves skewed distributions, such as a higher frequency of price increases compared to decreases in "Close" prices.

3. Mean Squared Error (MSE)

4. Mean Absolute Error (MAE)

5. R-squared (Coefficient of Determination)

3.2.Results:-

Performance Evaluation

1.Goal of the Method

Singular Value Decomposition (SVD) is used to extract trends from normalized stock data, and the level to which the reconstructed data closely resembles the original data is evaluated. Furthermore, stock movement is classified and trends are extended into the future using the reduced trends.

2.Quantitative Results

Reconstruction Accuracy:

Mean Squared Error (MSE): 0.0001

Mean Absolute Error (MAE): 0.0054

R-Squared: 0.9999

These results demonstrate how closely the reconstructed data resembles the actual data. A near-perfect R-squared score and extremely low MSE and MAE values indicate the SVD-based method's ability to successfully preserve the information in the original data.

Classification Performance:

F1 Score: 0.9549

After trained on the reduced trends, the logistic regression classifier does an excellent task of forecasting whether the stock will close higher than its opening price. The approach is flexible for this task, determined by its good precision and recall (F1 Score of 0.9549).

In conclusion, The technique validates the use of SVD for dimensionality reduction in financial time-series data by doing well in trend extraction and stock movement prediction.

In-Depth Performance Analysis:

1.Impact of Parameter k(Number of Singular Values Retained)

The total amount of primary components kept for trend analysis and reconstruction is determined by the parameter k. The trade-off between computational complexity and reconstruction accuracy is affected by changing k.

2.Experiment Setup:

The analysis is performed by reconstructing the data with different values of K(e.g., k=1,2,3,5,10) and the following metrics is calculated:

Reconstruction error (MSE, MAE, R-Squared).

Classification performance (F1 Score).

Execution time.

3.Results Table:

k (Singular Values Retained)	MSE	MAE	R-Squared	F1 Score	Execution Time (s)
1	0.0256	0.0342	0.9805	0.7123	0.02
2	0.0042	0.0104	0.9968	0.8431	0.03
3	0.0001	0.0060	0.9999	0.9297	0.05
5	0.0000	0.0023	1.0000	0.9332	0.07
10	0.0000	0.0010	1.0000	0.9345	0.10

Insights:

Reconstruction Accuracy: MSE and MAE rapidly decline as k rises, and R-squared gets closer to 1. But after k=5, the improvement is negligible.

Classification Performance: As k increases, the F1 Score gets better until it reaches a plateau at k=5.

Computational Cost: As execution time rises with k, smaller values of k are preferred for computing efficiency without considerably reducing accuracy.

4.Charts & Graphs

Reconstruction Metrics vs k: MSE, MAE, and R-squared are displayed as functions of k in a line graph.

F1 Score vs. k: An independent plot that shows the classification performance for different values of k.

Overall, In classification performance evaluation before SVD operation, The Accuracy score is 0.9508 and F1 Score is 0.9559.

After applying SVD operation, the Accuracy score is 0.9549 and F1 score is 0.9594

4.Individual Contribution:

Pooja: Wrote the Overview, explaining the problem, challenges with stock data, and the role of SVD in simplifying and analyzing trends.

Aishwarya: Prepared the Methodology, detailing data collection, normalization, SVD application, and evaluation steps.

Adithya and Keerthi Krishna Chand: Collaboratively worked on the implementation of code & Setup, focusing on data preparation (Adithya) and tools/environment setup (Keerthi Krishna Chand).

Vaishnavi: Handled the Results, analyzing reconstruction accuracy, evaluating classification performance before and after SVD, and the impact of k with metrics and graphs, calculating Accuracy score and F1 score etc..